PalmSource™ 99

Latest Graphics APIs: Lab Session

David Fedor
Senior Developer Support Engineer

3Com
More connected.

Schedule

PalmSource 99

- Move to the 3.5 SDK
- Test for, and fix some common bugs
- Switch to new Graphics APIs
- Add new features, or enhance existing ones

- Please ask questions as we go, and think about how this impacts your own applications

Moving to the 3.5 SDK

PalmSource 99

- Switch your build system to 3.5
  - The minimum is to copy the 3.5 support folder next to your 3.0 support folder
  - Other things are nice and good and helpful but not necessarily required
    - (See the readme!)
  - Also copy the new Palm OS® Emulator and the ROM images
- Convert your project to 3.5
  - Change the include path
  - #include <PalmOS.h>
  - Type changes

Project Conversion in Detail

PalmSource 99

- <Pilot.h> to <PalmOS.h>
- Change type usage in Signature.c
  (Use “Find Entire Word” and don’t “Ignore case”)
  - VoidHand to MemHandle
  - VoidPtr to void *
  - Ptr to either MemPtr or void *
  - DWord and ULong to UInt32
  - Word and UInt to UInt16
  - SWord and Int to Int16
  - (Other projects may need more changes)

Test for Common Bugs

- Test in the Palm OS Emulator with the new roms
- Check for common bugs:
  - Form contents not appearing if drawn before FrmDrawForm
  - Form contents not drawing on debug roms after an overlapping form has closed
    - This is to test your frmUpdateEvent handler
  - Various parameter checking alerts
  - Screen hacking not working any more
  - Memory leaks

Fix the Bugs in Signature.c

PalmSource 99

- Add code to StopApplication to free gBitmapPtr, and call FrmCloseAllForms()
  - While we're there, remove the MemPtrFree from GetSignatureBitmap and put it before the call to GetSignatureBitmap
- Call FrmDrawForm(theFormPtr) at start of MainViewInit, before drawing
- Call FrmDrawForm(theFormPtr) in the start of our frmUpdateEvent handler, and return true
- Remove the FrmDrawForm call in our frmOpenEvent handler

Switch to New Graphics APIs

PalmSource 99

- Add #define version35 0x03500000 and require that version in PilotMain()
- Remove code in GetSignatureBitmap, replace with calls to:
  - BmpCreate
  - WinCreateBitmapWindow
  - WinCopyRectangle
  - WinDeleteWindow
- Add code after the call to CaptureSignature():
  - WinPaintBitmap(gBitmapPtr, 0, 16);

Add New Features

- Draw border and contents by getting the right UI color
- Use WinPushDrawState
- Change the color of the user's "ink"
  - (This one isn't necessarily a good idea for a real application)
- Draw a one-bit bitmap in a shade for a background

PalmSource™ 99

Q&A

3Com
More connected.